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Flow boiling of nanofluids in microchannel heat exchangers: A critical review

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ABSTRACT

Nanofluids and microchannel heat exchangers have been studied extensively due to the advantages that using nanofluids can enhance the heat transfer performance for their larger surface area and the microchannel heat exchanger can decrease the coolant charge for their compact structures. The integration of the nanofluids into microchannel heat exchangers, therefore, has been considered an efficient strategy for cooling applications. However, this technology was still restricted by the small operating range, while the latent heat generated during the phase-change flow can partially improve the efficiency. This article critically reviewed some existing results and controversial issues relating to nanofluids and microchannel heat exchangers. Since the microchannel scale transition criteria have not been standardized yet, different transition criteria were summarised and compared in this article for the first time, as well as the influential factors that were emphasized for further consideration. Another reason for the debate is that flow boiling is highly sensitive, possibly leading to an inaccurate experiment result, thus several practically reasonable testing methods have been proposed to increase the accuracy of the experiment. In addition, the formulas for critical heat flux (CHF) and heat transfer rate empirical formulas were summarized, followed by the proposed future work.

1. Introduction

Humanity has become more dependent on cooling systems including radiators in cars, nuclear reactors, acoustic cavitation, heat pipes, and air conditioning systems [1–3], with the development of modern technologies. In cooling systems, one of the most challenging aspects is to effectively enhance the heat transfer performance [4,5]. Flow fluids and heat exchangers are two main factors that contribute to this [6,7]. Fig. 1 illustrates a complete development process for optimizing heat flow in a heat exchanger. Utilizing a pure coolant with a high latent heat capacity is one of the direct methods for improving the performance of the heat exchanger [8,9]. The identification of such a fluid was therefore a significant research objective before the 1980 s [10]. Due to the increased heat flux requirements of modern applications, especially electronic devices, pure fluids have reached their limits. Researchers have focused on nanofluids over the past few decades, which are the suspensions of well-dispersed nanoparticles in the base fluids [11–13].

Compared to pure fluids, nanofluids possess a greater heat transfer rate due to the large surface area of the nanoparticles in contact with the base fluid [14,15]. Furthermore, nanoparticles have a strong Brownian motion, which reduces the required amount of pumping power but provides high dispersion stability, thereby preventing clogging problems [16,17]. It is possible to adjust the properties of nanofluids by using different nanoparticles and base fluids. A variety of nanoparticles are used in nanofluids, including metal oxides (Al_2O_3, CuO, MgO, ZnO, SiO_2, Fe_2O_3, and TiO_2), metals (Cu, Ag, Ni, and Au), metal nitrides (AlN and SiN), metal carbides (SiC and TiC), semiconductors (TiO_2 and SiC), carbon materials (CNTs, MWCNTs, carbon nanotubes, diamond, and graphite), and other composites [18,19]. The most common base fluids are water, ethylene glycol, and oil. Therefore, nanofluid research has become a hot topic, which focuses on physical factors, force factors, and flow regime factors. Fig. 1 illustrates the physical characteristics of nanofluids, including nanoparticle diameter [20,21]; force factors are defined as the hydraulic forces between nanofluids and nanoparticles [22,23]; and flow regime factors are defined as the properties of nanofluids in various regimes [24,25]. Despite extensive research in nanotechnology, many critical issues still need to be resolved, such as mass production difficulties [21,26], limited operating ranges [20], and particle deposition problems [27,28].

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Nomenclature

D	Diameter, m
D_B	Diameter of the contacted zone, m
C_o	Confinement number
σ	Fluid surface tension coefficient, N/m
g	Acceleration of the gravity, m/s^2
ρ	Density, kg/m^3
B_o	Bound number
$Eö$	Eötvös number
L_{cap}	Laplace number
F_g	Buoyancy, N
F_σ	Surface tension force, N
F_d	Drag force, N
F_l	Inertial force, N
A	Area fraction
q_{tp}	Total heat flux, W/m^2
q_{cb}	Heat flux for convective heat transfer, W/m^2
q_{nb}	Heat flux for nucleate heat transfer, W/m^2
h_{tp}	Saturated flow heat transfer coefficient, W/(m^{2}K)
T_w	Channel wall temperature, K
T_s	Saturation temperature of the channel fluid, K
k_l	Fluid thermal conductivity, W/(mK)
k_{np}	Nanoparticles thermal conductivity, W/(mK)
Nu	Nusselt number
Re	Reynolds number
Pr	Prandtl number
u	Relative velocity, m/s
μ_∞	Fluid viscosity at free stream temperature, Pa·S
μ_w	Wall superheat temperature, K
P_H	Wetted perimeter of a channel, m
P_L	Heated perimeter of a channel, m
We	Weber number
Na	Active nucleate site density
X_{tt}	Lockhart-Martinelli parameter
G	Mass flux, kg/(m^{2}s)
H_{lv}	Fluid heat latent, J/kg
r^*	Critical radius of cavity, m

Acronyms

OBE	Onset of bubble elongation
CHF	Critical heat flux
ONB	Onset of nucleate boiling
OFI	Onset of flow instabilities

Greeks

φ	Nanoparticles concentration
μ	Viscosity, Pa·S
σ	Interfacial tension, mN/m
Θ	Contact angle, °

Subscripts

h	Hydraulic
L	Liquid
V	Vapor
crit	Critical
th	Threshold
np	Nanoparticles
l	Base fluid
fc	Forced convection
nb	Nucleate boiling
Sat	Saturated
A	Apparent
D	Departure
in	Condition at inlet

The performance of heat transfer can also be enhanced through novel design of heat exchangers. Several factors can be considered: 1) Microelectronics are becoming increasingly popular, which requires smaller geometries and lower coolant flows for the cooling devices to be more compact; 2) The price of devices is declining, leading to the need for more cost-effective cooling systems [29]; 3) There are several points of concern resulting from the trend towards increasing packaging density, such as high heat dissipation and low local surface temperatures [30]. In response to these demands, Tuckerman and Pease at Stanford University firstly proposed the concept of microchannel heat exchangers in the early 1980 s [31].

With the development of electronic technology, the thermal management for microchannel heat exchangers has been with more requirements [32–34]. The main objectives of research regarding microchannel heat exchangers are to improve heat transfer efficiency and to solve problems associated with non-uniform heat flux [35,36]. The modification of the microchannel structures, on the one hand, can address these problems, resulting in a greater ratio of heat transfer area to volume. Fig. 1 illustrates some common changing methods, such as rarefaction, flow disruption, and re-entrant microchannels. Specific microchannels available on the market are the pin fin microchannel, the wavy fin microchannel, the oblique fin microchannel, and the double-layered microchannel [37–39]. In addition, surface conditions can also affect the properties of microchannel heat exchangers [40,41].

The combination of nanofluids and microchannel heat exchangers can enhance heat transfer rates [42–44]. It was initially found that simple non-phase change nanofluids could enhance heat transfer and reduce pressure drop in microchannel heat exchangers [45–47]. Nevertheless, the small operating range has restricted the nanofluid applications in industrial settings. It was only possible to use nanofluids below the boiling point. Meanwhile, latent heat was not fully utilized. For the purpose of solving these problems, phase change nanofluids have been studied in nanotechnologies, including pool boiling and flow boiling [48–50]. Few researchers attempted to investigate the flow boiling of nanofluids before 2010. Over time, researchers began to realize the importance of multi-phase nanofluids [51]. Recent years have seen an increase in interest in this topic.

Most existing review articles just have focused on single-phase heat transfer or pool boiling heat transfer [37,41,52,53], and very few articles exclusively address and consider flow boiling in microchannel heat exchangers. Due to the challenges of small operating ranges in nanofluid applications, it is necessary to pay more attention to the flow boiling of nanofluids. Also, the existing review articles only focused on the parameters and the phenomenon of heat transfer coefficient, convection heat flux, pressure drop, nanoparticle stability, flow pattern, and bubble dynamics [10,14,54]. But no recent articles have summarized the microchannel scale transition criteria or explained the controversy surrounding the experiments and simulations. In addition, this article summarizes the flow boiling characteristics of nanofluids in microchannel heat exchangers. Several inconsistent conclusions among researchers were highlighted, and some reasonable suggestions for future work were made. There were five sections in this review paper: *Section 1* introduces the development process; flow boiling and microchannel heat exchangers are described in *Section 2*, focusing on channel classification standards; *Section 3* summarizes some evaluation indices for flow boiling of nanofluids in microchannel heat exchangers and discusses controversial issues; In *Section 4*, the results from experiment results and numerical models are compared and analysed; a summary of the paper is provided in *Section 5*, followed by the future path based on the existing study.

2. Flow boiling in microchannel heat exchangers

Cooling with a phase change fluid in a microchannel heat exchanger dissipates a large amount of heat flux while maintaining the same flow rate, which is one of the most efficient methods for improving heat transfer rates. It has been demonstrated that flow boiling in microchannel heat exchangers uses a small amount of coolant to produce a uniform wall temperature due to their saturation properties [16,55].

2.1. Definition of flow boiling

According to the flow dynamics, phase change heat transfer can be divided into two categories: pool boiling and flow boiling (forced convection boiling). In comparison with pool boiling, flow boiling is a more complex research topic. Since the boiling process occurs in the freedom surface for pool boiling, the bubbles undergo a complete growth process and are predictable. For flow boiling, the microchannel restricts the growth of bubbles, so a variety of factors affect bubble growth. There have been several investigations into the mechanisms and heat transfer correlations involved in pool boiling over the past few decades. However, flow boiling research is still at the initial stage, and there are some controversies among researchers.

As shown in Fig. 2, flow boiling can be classified as subcooled or saturated boiling, depending on the temperature of the working fluid [56]. The process of subcooled boiling occurs at a heated wall when the bulk average temperature is less than the saturation temperature; saturated boiling occurs when the bulk average temperature is at the saturation temperature. In the present day, most studies are concerned with subcooled boiling [56]. Due to the sensitivity of saturated flow boiling and the difficulty in controlling, the same experimental setup can produce different experimental results and conclusions. Due to this, different results have been reported in the literature regarding the characteristics of nanofluids flowing through microchannel heat exchangers. In the following sections, we will discuss these differences and future research directions.

2.2. Description of microchannel heat exchanger sizes

Microchannel heat exchangers are widely used in modern industry due to their compact structures and process intensification. Both

Fig. 1. Complete development process for nanofluids and microchannel heat exchangers.

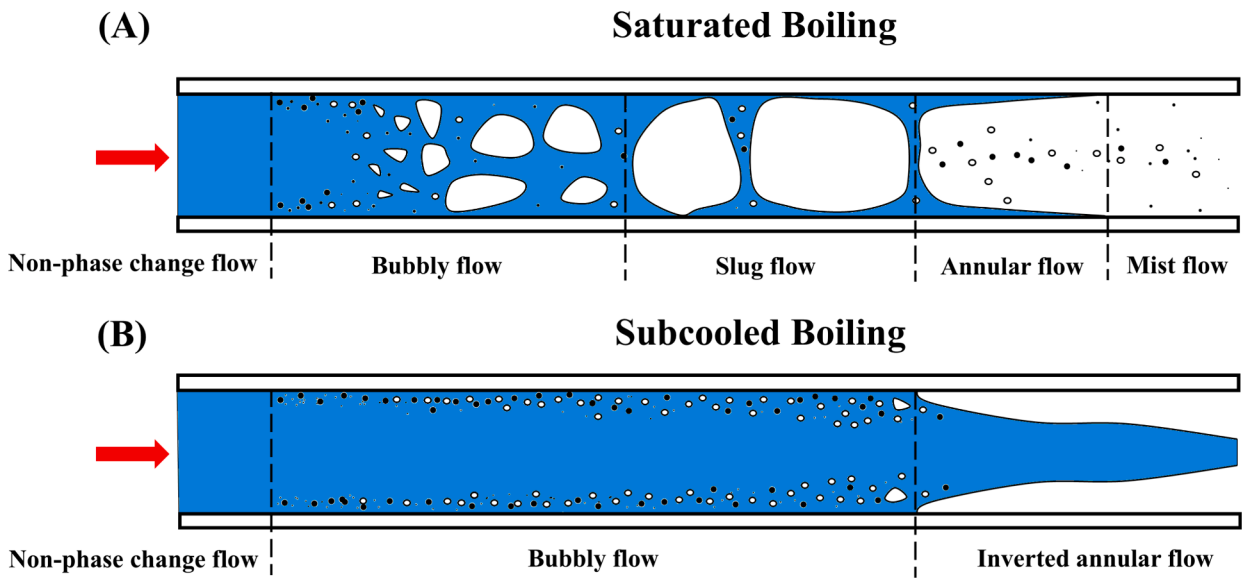

Fig. 2. Microchannel flow boiling can be classified into two types: (A) saturated boiling and (B) subcooled boiling.

economic and ecological benefits have been derived from these characteristics. As modern electronics have developed, it has become necessary to develop highly efficient dissipation cooling applications. A small channel cooling system provides efficient heat exchange and is comprised of three types: microchannels, minichannels, and macrochannels. It is important to note that different channel sizes have different effects on heat transfer, especially when it comes to phase change flow. Several characteristics and heat transfer phenomena are suppressed or disappear as channel sizes increase. The inconsistent selection of microchannel heat exchanger sizes used in the experiments also contribute to the controversy results.

It is necessary to determine how the channel size should be defined [57]. The easiest method of classifying them is by using their hydraulic diameters. Table 1 provides a list of different classification criteria. There are several shortcomings to this classification, including that it does not take the fluid properties and the heat transfer characteristics or mechanisms into account. Phase change fluids exhibit different flow patterns, pressure gradients, and heat transfer coefficients depending on the channel size. Consideration should be given to surface roughness and fluid properties. Therefore, this classification method is only applicable to non-phase change flow systems.

In the case of a phase change flow, the transition criteria become more complex. There is also a difference between the criteria for boiling flow and condensation flow. The following section will introduce some dimensionless numbers that are used to classify channel sizes. Nowadays, dimensionless numbers are widely used, but the definition of a critical value is subject to considerable debate.

The confinement number (C_o) is the most commonly used dimensionless number to classify channels according to the situation in which bubbles are confined within the channel, and is given by:

$$C_o = \frac{1}{D_h}\sqrt{\frac{\sigma}{g(\rho_L - \rho_V)}} \tag{1}$$

Table 1
Macro- to micro-scale transition criteria based on hydraulic diameters.

Research	Microchannel	Minichannel	Macrochannel
Mehendale et al. (2000) [58]	0.001–0.1 mm	0.1–1 mm	>1 mm
Kandlikar & Grande (2002 & 2003) [59,60]	0.05–0.6 mm	0.6–3 mm	>3 mm

Where σ is the fluid surface tension coefficient; D_h is the hydraulic diameter; ρ_L is the liquid density; and ρ_V is the vapor density. However, different thresholds have been identified for boiling flow, as shown in Table 2. The reason for the different thresholds is primarily influenced by the fluid properties. In addition, some researchers believe that the thresholds for condensation flow are different from those for boiling flow, as shown in Table 3. However, there is a limited amount of research on this aspect, and this conclusion can only be verified for cooling systems. It is worth noting that Li and Wang [61] introduced the concept of the transition region. Currently, there are no effective methods for calculating or testing this region.

Another typical dimensionless number used to evaluate channels is the bound number (B_o), which represents the ratio of gravity to surface tension force [64], as follows:

$$B_o = \frac{g(\rho_L - \rho_V)D_h^2}{\sigma} \tag{2}$$

Brauner and Ullmann [64] provided a threshold for the Bound number as $B_o < 0.05$ to define a microchannel where gravity can be ignored; $0.05 < B_o < 3$ to define a minichannel where gravity is less than the surface tension force; and $B_o > 3$ to define a macrochannel because the surface tension is higher than gravity. Apart from the Bound number, Brauner and Ullmann [64] also proposed an E ö tv ö s number ($E\ddot{o}$) based on the transition of flow patterns, which is given by:

$$E\ddot{o} = \frac{g(\rho_L - \rho_V)D_h^2}{8\sigma} \tag{3}$$

This number is relevant to the liquid films and the wetting ability of the channels. The evaluation criteria are similar to those for the bound number, i.e. $E\ddot{o} < 0.2$ is a microchannel, and $E\ddot{o} > 0.2$ is a macrochannel. Comparing the three dimensionless numbers mentioned above, Ong and

Table 2
Transition criteria for boiling flow.

Research	Microchannel	Minichannel	Macrochannel
Kew & Cornwell (1997) [62]	$C_o > 0.5$	$C_o < 0.5$	
Triplett et al. (1999) [63]	$C_o > 1$	$C_o < 1$	
Brauner & Uiimann (2006) [64]	$C_o > 0.79$	$C_o < 0.79$	
Ong & Thome (2011) [65]	$C_o > 1$	$0.3-0.4 < C_o < 1$	$C_o < 0.3-0.4$

Table 3
Transition criteria for condensation flow.

Research	Microchannel	Minichannel	Macrochannel
Li & Wang (2003) [61]	$C_o > 4.46$	$0.57 < C_o < 4.46$	$C_o < 0.57$
Cheng & Wu (2006) [66]	$C_o > 4.472$	$0.577 < C_o < 4.472$	$C_o < 0.577$

Thome [65] proposed a new correlation, as follows:

$$E\ddot{o} = \frac{B_o}{8} = \frac{1}{8{C_o}^2} \tag{4}$$

Additionally, some optimization methods for these three dimensionless numbers have also been proposed for enhancing the consistency of the transition criteria method based on the actual situation [67–69].

Apart from the fixed critical and threshold values, there is also another method for classifying channels. According to Suo and Griffith [70], experiments with capillary tubes demonstrated that the Laplace number (L_{cap}) represents the relationship between gravity and surface tension force, and is given by:

$$L_{cap} = \sqrt{\frac{\sigma}{g(\rho_L - \rho_V)}} \tag{5}$$

Li and Wang [61] further investigated the Laplace number and built correlation formulas for critical diameter (D_{crit}) and threshold diameter (D_{th}) by this dimensionless number, as follows:

$$D_{crit} = 0.224L_{cap} \tag{6}$$

$$D_{th} = 1.75L_{cap} \tag{7}$$

Therefore, channels are categorized according to the critical diameter and threshold diameter of their internal tubes. $D < D_{crit}$ defines as a microchannel where gravity is insignificant compared to surface tension force; $D_{crit} < D < D_{th}$ defines as a minichannel where gravity equals to surface tension force; and $D > D_{th}$ defines as a macrochannel where surface tension has the predominant effect over gravity [71].

As a summary, for non-phase change flows, the macrochannel to microchannel transition criteria are determined directly by the hydraulic diameter. For phase change flows, the transition criteria between a macrochannel and a microchannel can be roughly interpreted as that the capillary length is approximately equal to the hydraulic diameter, which implies that bubbles are similar to the cross-section area of the channel.

3. Evaluation indexes for flow boiling of nanofluids in microchannels

It is necessary to put forward some evaluation indexes since the performance improvement of flow boiling can be observed. One of the most important parameters is critical heat flux (CHF), which is greatly influenced by nanofluids, and also impacted by other evaluation indexes, such as the onset of nucleate boiling (ONB), the onset of bubble elongation (OBE), and the onset of flow instability (OFI).

3.1. Development of critical heat flux

CHF is an essential criterion for evaluating phase change flow, which determines the safety of its usage. The heat flux operating conditions must be below the CHF. When the heat flux exceeds the value, the surface temperature raises immediately, resulting in a sudden decrease in heat transfer efficiency. This occurs due to the formation of a vapor blanket and the subsequent drying out of the heating surface, resulting in a lack of fluid and the destruction of the device. The burnout phenomenon occurs in subcooled and saturated boiling conditions [55]. Thus, increasing CHF is important for protecting devices, particularly for small channel systems.

CHF has received increased attention from researchers since the 1950 s; consequently, numerous prediction correlations of CHF have been proposed [54,72–74]. Based on these correlations, the mechanism of CHF on water flow in the tube has been studied [75]. As nanotechnology developed, You et al. [76] became pioneers in the study of the effect of CHF on nanofluids and determined that nanofluids increased CHF under pool boiling conditions. With increased boiling time, Okawa et al. [77] concluded that static contact angle and CHF asymptotically decreased and increased, respectively, due to surface wettability in pool boiling. Park et al. [78] tested different heating orientations for pool boiling and reported that vertical orientation enhanced CHF limits up to about 40 % and horizontal orientation about 200 % at maximum. Gradually, Kim et al. [79] began to study the CHF of nanofluids under flow boiling, marking a turning point in the field of CHF.

As shown in Fig. 3, CHF increased with the trend of mass flow. It was demonstrated by Roday et al. [80] that CHF increased with an increase in both mass flow and coolant exit pressure. Vafaei and Wen [55] found that CHF increased linearly with the trend of mass flow rate at different volume fractions of nanofluids. Furthermore, Tibirica et al. [81] found that CHF grew with the trend of mass flow in all types of microchannel shapes and that CHF gradients decreased with the increasing mass flow compared to low mass flows. In the case of low mass flow, CHF is associated with wall dry-out as a result of a lack of liquid film on the channel surface. Nevertheless, in the case of high mass flow, the enhancement of interfacial shear and entrained liquid forms a layer of liquid film. When the mass flow increases, the liquid film becomes thinner, concurrently, increasing CHF. According to Do Nascimento et al. [82], nanofluids and pure fluid both exhibited an increase in CHF when the mass flow rate was increased.

In contrast to mass flow, there are no exact conclusions regarding the effect of subcooling on CHF. Roday et al. [80] found that the changing trends of CHF were different within the ranges of subcooling, as shown in Fig. 4. CHF decreased within a low subcooling range, and vice versa. The void fraction was very high at low subcooling, which increased the average velocity, resulting in a high CHF value. In addition, they believed that it was very difficult to detect trends in CHF that were influenced by exit quality. Vafaei and Wen [83] concluded that CHF increased with the increase of subcooling. It should be noted, however, that they compared only the subcooling temperature at 45 °C and 80 °C, so the conclusion has some limitations. There is currently no comprehensive understanding of the effect of subcooling on CHF. In future experiments, more influence factors should be considered (e.g. different nanoparticle types, nanoparticle concentrations, and subcooling temperatures).

3.2. Influence of nanoparticles on CHF

In the study of normal channels, nanofluids have been shown to increase CHF due to their surface wettability [84–87]. Ahn and Kang et al. [88,89] developed a new flow boiling model based on the wetting-zone fraction. However, the model did not take the mechanism of surface wettability into account. In other words, this conclusion can only be applied in regular channels but not the nanofluid flow boiling in microchannel heat exchangers. As a result, several new mechanisms of CHF changing were developed, including: (1) Surface wettability (2) Bubble dynamic theory (3) Active nucleation sites (4) Dry-out behaviour.

3.2.1. Surface wettability

An explanation for the influence of nanoparticles on CHF can be attributed to the existence of structure disjoining pressure. Wasan and Nikolov [90] discovered that structural disjoining occurred in nanofluids, not in pure fluids. This was attributed to the Van der Waals force between nanoparticles. It was further elaborated by Wen [91] and Xu [92] that the disjoining pressure caused a change in the three-phase contact line, reducing the contact area between the dry vapour and

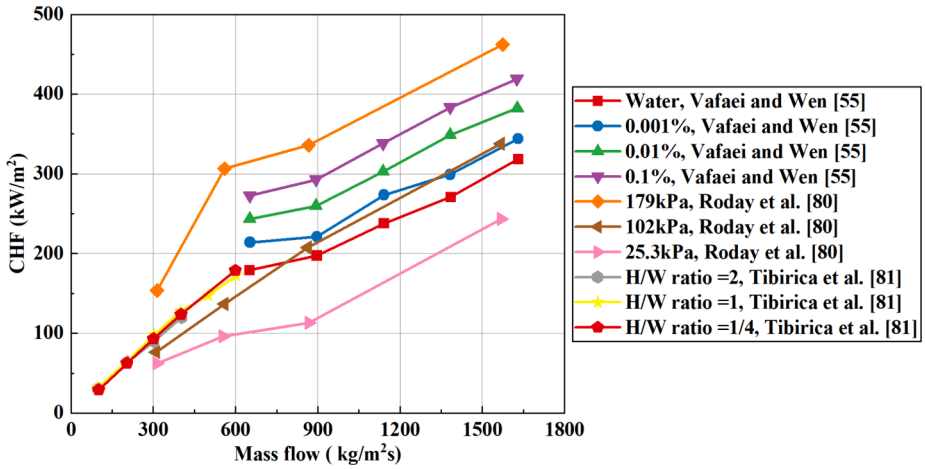

Fig. 3. CHF varies with mass flow.

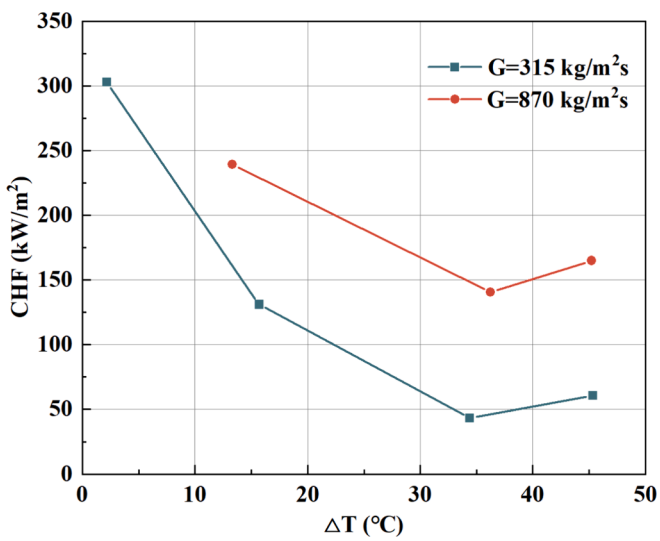

Fig. 4. CHF varies with inlet subcooling [80].

the heating surface. In Fig. 5 [92], (a) is the dry vapour directly heated on the solid surface, (b) is the thin liquid film between the dry vapour and solid heating surface, (c) is the liquid or nanofluid area, and (d) is the thin liquid film with nanoparticles. The structural disjoining pressure is responsible for forming (d). Initially, some nanoparticles are deposited in the local area. After that, the force interaction between nanoparticles becomes stronger, causing particles to accumulate. Upon clustering, the nanoparticles would gather in the depth direction, forming the second film layer. In this regard, nanofluids can improve the wettability of a heating surface, thereby altering the coefficient of heat transfer.

During the experiment, Kim et al. [93] observed that the nanoparticles formed a porous layer on the heating surface. At the same heat flux, nanoparticles improved the surface wettability and retained more bubbles on the heating surface. Due to the reduced contact angle on the heating surface, adding nanoparticles enhanced CHF limitation. Thereafter, Kim et al. [94] investigated flow boiling by comparing alumina-water nanofluid with water flow boiling. In the experiments, the channel with running water was completely burnt out, while the channel with nanofluid had only localized burnout, indicating that nanoparticles increased the CHF value. Next, Kim et al. [79] conducted the same experiments using different nanoparticles in different mass fluxes. By adding nanoparticles, CHF was enhanced by 40–45 % at high mass fluxes (2000–2500 kg/m^2 • s), and a similar CHF limitation was obtained at a low mass flux (1500 kg/m^2 • s). Vafaei and Wen [55] demonstrated that CHF increased nonlinearly with particle concentration. Several experiments have shown that the CHF of alumina-water nanofluid is 4–30 % higher than that of deionized water, and the value is higher at low mass fluxes [95].

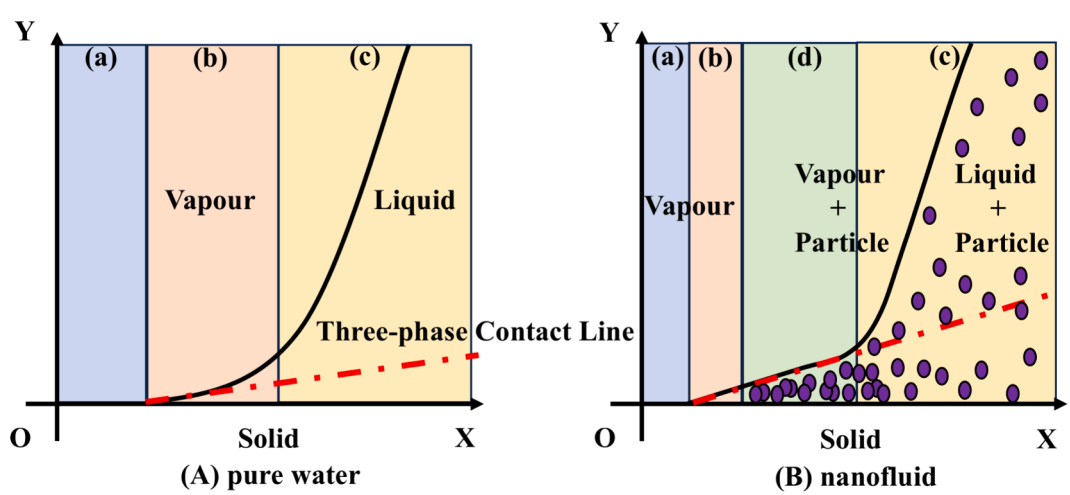

Fig. 5. Flow boiling mechanisms.

3.2.2. Bubble dynamic theory

The nanoparticles affect the formation of bubbles and the frequency of bubble formation on the boiling surface. According to the bubble dynamic theory, bubble growth and departure are controlled by buoyancy F_g, surface tension force F_σ, drag force F_d and inertial force F_l, as shown in Fig. 6 [92]. The specific balance equation is illustrated by:

$$\vec{F_\sigma} + \vec{F_l} = \vec{F_g} + \vec{F_d} \tag{8}$$

Nanoparticles have a reduced diameter of the contacted zone D_B due to the decreased structural disjoining pressure. In accordance with Eq. (9), the surface tension force is reduced, so the bubble size of the nanofluid becomes smaller compared to that of pure water before departing [92], as follows:

$$F_\sigma = \sigma D_B \pi sin\theta \tag{9}$$

The occurrence of local dry-out when bubbles exit this area can be weakened by reducing the bubble growth time. As a result, nanofluids have a higher CHF value than pure fluids.

3.2.3. Active nucleation sites

For a more detailed understanding of the relationship between the boiling surface and nanoparticles, cavities on the heating surface were also considered. Yu and Sur et al. [96] concluded that nanoparticles filled the cavities on the boiling surface as a result of nanoparticle deposition. As shown in Fig. 7(A), nanoparticles can reduce nucleation sites to stabilize the boiling process. Nevertheless, Narayan and Anoop et al. [97] held a different opinion and believed that the number of nucleation sites was determined by the ratio of roughness to particle size. Based on Fig. 7 (B), if the cavities were small, the deposited nanoparticles would form new nucleation sites above the boiling surface, which would accelerate the boiling process. Currently, there is no consensus opinion and no quantitative evidence supporting the theory.

Yu and Sur et al. [96] offered another explanation, that the thermal boundary layer could affect active nucleation sites. Hsu [98] proposed that the thickness of the thermal boundary layer was related to the nucleation sites. Additionally, single-phase heat transfer research revealed that nanofluids suppress the development of thermal boundary layers as a result of their viscosity and thermal conductivity [99–101]. As a result, Yu and Sur et al. inferred the results indirectly. At present, there is no effective experiment result that verifies the conclusions.

3.2.4. Dry-out behaviour

Do Nascimento and Moreira et al. [82] offered a different interpretation of CHF changes. They suggested that the phenomenon should be analysed separately for subcooled and saturated nanofluids in order to determine the cause. Adding nanoparticles to subcooled (low vapour quality) nanofluids increased their CHF due to the nucleation process and surface wettability/wickability. However, the CHF changing mechanism in saturated (high vapour quality) nanofluids was due to the dry-out mechanism. The results of the experiments demonstrated that nanoparticle diameter, concentration, and composition did not influence CHF, therefore surface wettability did not influence it. It is actually the periodic supply of liquid lumps that is the primary cause [102–104]. A rise in liquid temperature caused liquid films to evaporate. Afterwards, the liquid lumps attempted to resupply. As the liquid temperature increased, evaporation accelerated and the liquid lumps were unable to supply in time, resulting in an increase in CHF.

3.2.5. Reasons for inconsistencies

Based on the reviews above, Table 4 summarizes the effects of nanoparticles on CHF. The CHF changes can be explained by four mechanisms: (1) Flow boiling nanofluids exist structure disjoining pressure, forming a porous layer on the surface that increases the surface wettability. (2) According to bubble dynamic theory, nanoparticles increase the CHF value by altering the bubble formation and formation frequency on the boiling surface. (3) The surface condition and the thermal boundary layer have an effect on active nucleation sites, resulting in a change in CHF. (4) Liquid lumps are unable to provide the evaporation of liquid in time, therefore the CHF value increases.

The following three factors are responsible for the different opinions regarding the varying trend of the CHF: (1) As flow boiling is very sensitive, it was possible to obtain different experimental results with the same setup. (2) Measurement methods have errors. As a result of technical limitations, when thermocouples adhere outside the channel wall directly to test temperatures, the thermal inertia of the channel wall and thermocouple hot junction affects the temperature results. Meanwhile, it is difficult to measure the periodic changes in temperature of the liquid lumps. (3) Measurement uncertainty is also caused by the oscillations of the system. As a result of bubble growth in the confined channel, thermal instabilities and backflows are difficult to observe.

3.3. Other evaluation indexes

3.3.1. Onset of nucleate boiling

ONB marks the transition from natural convection to nucleate boiling. The boiling instabilities in microchannels can be suppressed and delayed through several methods: first, changing the surface properties of the microchannels; secondly, adding jets to microchannel heat exchangers can result in severe disruption of the flow stream and development of thermal and hydrodynamic boundary layers near the channel wall [30]; and finally, nanofluids can be used in microchannels.

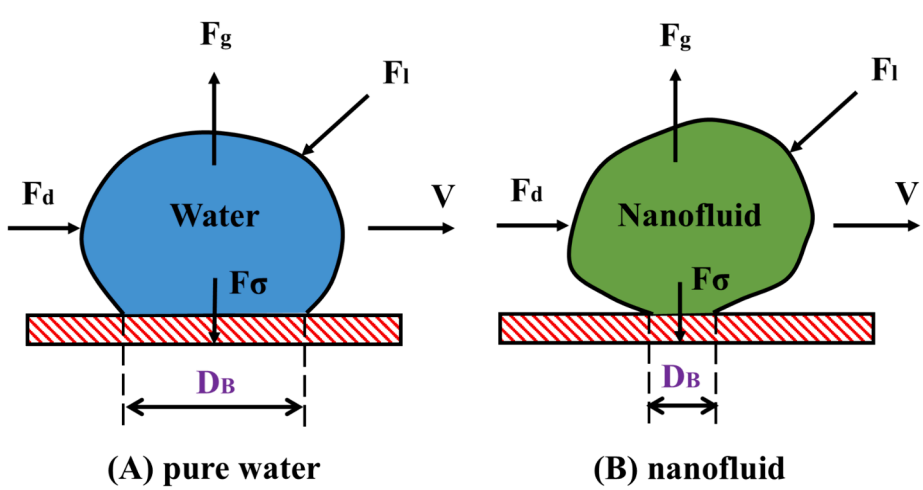

Fig. 6. Force balance for growing bubbles in two situations.

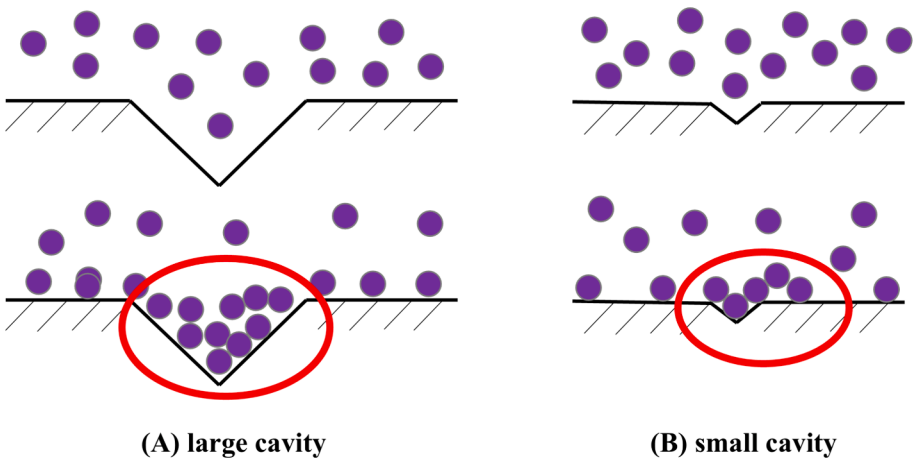

Fig. 7. The nucleation sites before and after deposition.

Table 4
Summary of the influence of nanoparticles on CHF.

Reasons	Influence of Nanoparticles on CHF	Explanations
Surface wettability	Increasing CHF	Nanoparticles change the three-phase contact line, reducing the contact area between the dry vapour and the heating surface.
Bubble dynamic theory	Increasing CHF	The nanoparticles reduce the diameter of the contacted zone, which affects the formation of bubbles.
Active nucleation sites	Inconsistency	1. Nanoparticles fill the cavities on the boiling surface as a result of increasing CHF. 2. The deposited nanoparticles form new nucleation sites above the boiling surface as a result of the reduction in CHF. 3. Active nucleation sites are also influenced by the thermal boundary layer, so the influence cannot be determined.
Dry-out behaviour	Increasing CHF	Evaporation accelerates as liquid temperature increases, and the liquid lumps are unable to supply the liquid in time, resulting in an increase in CHF.

Soleimani et al. [48] simulated highly subcooled flow boiling in a microchannel heat sink to examine the effects of different concentrations of alumina nanoparticles on HFE-7100, with Reynolds numbers ranging from 530 to 2000, and boiling numbers ranging from 0.0023 to 0.0071. They found that the wall temperature decreased as nanoparticle concentration increased, so the reduction in wall temperature delayed ONB. Yu and Sur et al. [96] experimentally investigated the forced convective flow boiling and two-phase flow of Al2O3–water nanofluids in a minichannel, and the characteristics of flow boiling heat transfer as well as the ONB of nanofluids were measured. They concluded that nanofluids could delay ONB because of the surface conditions and wettability of the fluids.

3.3.2. *Onset of bubble elongation*

Due to the confinement of the bubble growth, OBE occurs most commonly in microchannels. According to Fig. 8, the data are from Edel and Mukherjee [105], who evaluated the location of OBE at a Reynold number of 300 and a heat flux of 130 kW/m^2. There was a random location of the water on the OBE, while the nanofluids were located in a regular pattern. This was due to the fact that nucleation sites were difficult to form in the water, so nucleation site locations were unpredictable. Moreover, nanofluid regulation became stronger over time, and OBE occurred slowly with an increase in deposition caused by wicking.

3.3.3. *Onset of flow instability*

OFI is the point at which flow boiling begins to cause flow instabilities. Instability caused by flow boiling can be divided into static instabilities and dynamic instabilities, which have a variety of disadvantages [106]. Heat transfer can be hindered by oscillations of flow rate, pressure and temperature; the oscillation of the temperature may also produce hotpots, which may adversely affect the reliability and safety of microchannel heat exchangers [107,108].

4. Experiments and simulations

Flow boiling experiments for nanofluids in microchannel heat exchangers are still in the early stages of exploration. The majority of experimental setups are similar to Fig. 9, with the specific test sections being determined by the purpose of the experiment. Table 5 summarizes the relative results of published papers in the past 15 years. Several important factors have been listed, which can help understand the development process. The coolants used in these published papers mainly consisted of Al_2O_3 and pure water. The Al_2O_3 nanoparticles possess uniform diameter distribution, high resistance, and good sound insulation properties. Flow boiling experiments require precise control of surface temperature. As a result of using water as a coolant, the peak surface temperature was approximately 150-200°C [55]. As far as simulation research is concerned, most of the simulation correlations are fit by the results of the experiments. It is difficult to draw direct correlations from the flow boiling mechanism due to a lack of a clear understanding of it. In the following section, some important experiments and simulation results will be discussed.

4.1. *Temperature fluctuation*

Nanofluids can reduce temperature fluctuations in microchannels. As a result of the reverse flow for flow boiling in the microchannel, the inlet temperature varied cyclically [109]. A reduction in compressibility of the inlet flow and an increase in mass flow are two ways to reduce the inlet temperature fluctuations. A fluid with high compressibility is not degassed [110], and the liquid at the inlet is pushed back by growing confined bubbles, causing pressure and temperature fluctuations [111]. It has been shown that nanoparticles can increase the mass flow and reduce the compressibility of fluids. Consequently, Vafaei and Wen [55] observed that nanofluids containing 0.01 vol% and 0.1 vol% alumina maintained a constant inlet temperature.

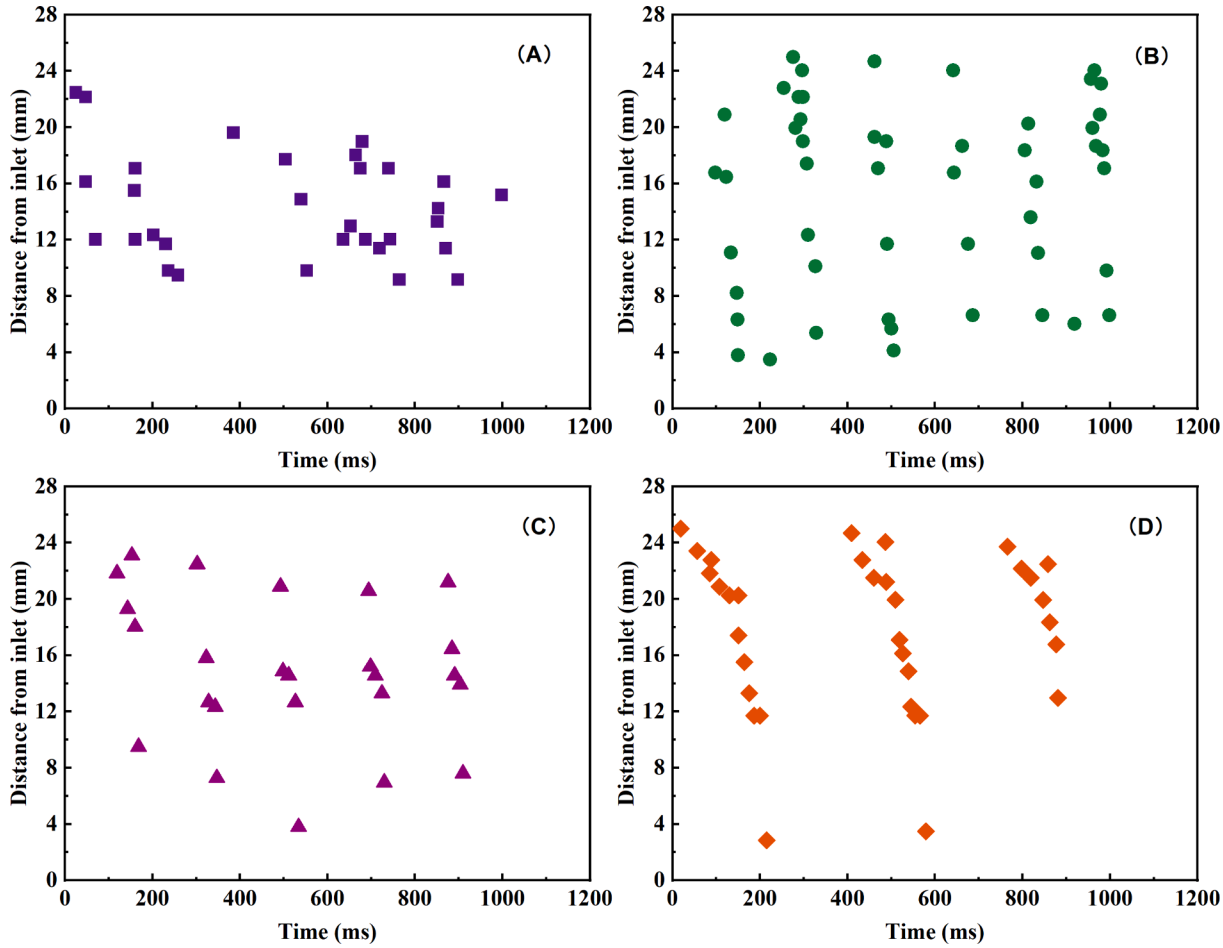

Fig. 8. OBE location as measured from the inlet for a heat flux of 130 kW/m^2: (A) water; (B) nanofluid, 25 mins; (C) nanofluid, 75 mins; and (D) nanofluid, 125 mins.

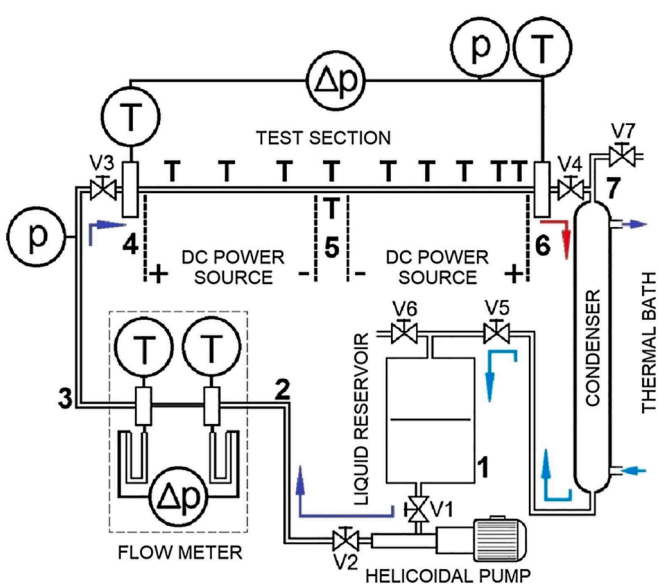

Fig. 9. Schematic of the experimental apparatus [82]. (Reprinted with permission from Elsevier).

Apart from reducing fluctuations in the inlet temperature, nanofluids can also reduce fluctuations in the average temperature. Through experiments, Yu and Sur et al. [96] demonstrated that nanofluids improved two-phase flow stability. As shown in Fig. 10, data from Yu and Sur et al. [96] found that increasing the nanoparticles brought the OFI boundary line closer to the single-phase line (heat fluxq_w = 0 W/cm^2), which meant that the nanofluids behaved like single-phase fluids. This is caused by nanoparticles reducing the size of the bubble and delaying the progress of the bubble's production. This results in a reduction in the compressibility of nanofluids, as well as a suppression of oscillations. In addition, Vafaei and Wen [55,83] observed that the temperature and pressure fluctuated at the inlet and outlet of the channel. In the confined channel, bubble nucleation and growth behaviour resulted in smaller oscillations of nanofluids than pure fluids. According to Duursma and Sefiane et al. [112] periodic oscillations occurred during the boiling process, and the amplitudes of the inlet and outlet pressures were higher than those of the inlet pressure.

4.2. Heat transfer rate

In terms of system evaluation, heat transfer rate is one of the most important factors. It has been reported that nanofluids increase heat transfer rates. However, the majority of these statistical studies are still based on simulations rather than on reliable experimental data, and there is no standard evaluation to determine whether nanofluids can improve heat transfer. Chehade et al. [115] found that nanofluids significantly increased heat transfer at low concentrations. Local heat transfer was highest at the microchannel entrance and decreased along the flow direction as the vapour quality changed. Soleimani et al. [48] concluded that adding nanoparticles could increase heat transfer using the volume of fluid (VOF) to simulate flow boiling. However, the increasing trend of single-phase boiling flow was less significant than that of two-phase flow boiling. By adding nanoparticles Al_2O_3 to water,

Table 5
Compare the results of experiments and simulations in the past 15 years.

Researchers	Working Fluids	Channel Shape	Channel Size	Heating Conditions	Analyse Methods	Main findings and/or results
Kim & McKrell (2010) [113]	Pure water, Al_2O_3-water (40 nm, 0.001, 0.01 and 0.1 vol%), ZnO-water (77.4 nm, 0.001, 0.01 and 0.1 vol%), and Diamond-water (165.4 nm, 0.001, 0.01 and 0.1 vol%)	Stainless steel tube	Wall thickness: 0.41 mm Internal diameter: 6.35 mm Length: 100 mm	Inlet temperature of 80°C	Experiment & Simulation	A simulation model to predict the changes in the density of ensuing nucleation sites. Flow boiling CHF was enhanced by 50 %.
Boudouh & Gualous et al. (2010) [114]	Pure water and CuO-water (35 nm, 5, 10 and 50 mg/L)	Rectangle channel	Hydraulic diameter: 0.8 mm	Fluid temperature of 100°C	Experiment & Simulation	Compared with pure fluids at the same mass flux, nanofluids showed higher local heat transfer coefficients, higher local heat fluxes, higher pressure drops, and lower surface temperatures.
Vafaei & Wen (2010) [55]	De-ionized Water and Al_2O_3-water (25 nm, 0.001, 0.01 and 0.1 vol%)	Stainless steel tube	Wall thickness: 0.16 mm Internal diameter: 0.51 mm Length: 306 mm	Fluid temperature up to 150°C	Experiment	CHF increased by 51 % at low nanoparticle concentrations.
Vafaei & Wen (2011) [95]	De-ionized Water and Al_2O_3-water (25 nm, 0.001, 0.01 and 0.1 vol%)	Stainless steel tube	Wall thickness: 0.16 mm Internal diameter: 0.51 mm Length: 306 mm	Fluid temperature up to 150°C	Experiment	Compared with deionized water, CHF was modestly increased with very low particle concentrations, i.e., 31 % with 0.1 % nanoparticles.
Ahn & Kang et al. (2011) [89]	Al_2O_3 nanoparticles coated surface (0.01 vol%)	Rectangle channel	Width: 10 mm Height: 5 mm	Two heat fluxes of 1200 kW/m^2 and 2344 kW/m^2	Experiment & Simulation	When the flow velocity was changed from 0 − 4 m/s, water boiling on nanoparticle-coated heaters was conducted and the CHF increased at a given velocity.
Ahn & Kang et al. (2012) [88]	Al_2O_3 nanoparticles coated surface (0.01 vol%)	Rectangle channel	Width: 10 mm Height: 5 mm	Heat fluxes ranged from 600 kW/m^2 to 2344 kW/m^2	Experiment & Simulation	A new flow boiling model was developed based on the wetting-zone fraction.
Xu & Xu (2012) [92]	Pure water and Al_2O_3-water (40 nm, 0.2 vol%)	Rectangle channel	Length: 7.5 mm Width: 0.1 mm Height: 0.25 mm	Around 100°C	Experiment	In a timescale of 100 s, elongated bubbles alternate with miniature bubbles. Nanofluids could reduce pressure drop and increase heat transfer in the flow boiling.
Chehade et al. (2013) [115]	Pure water and Ag-water (25 and 50 mg/L)	Rectangle channel	Hydraulic diameter: 0.8 mm	Inlet temperature of 60°C and supplied power of 200 W	Experiment & Simulation	Local heat transfer coefficients can be enhanced by 86 % and 200 %, respectively. Low nanoparticle concentrations exhibited higher heat transfer rates than high nanoparticle concentrations.
Morshed & Paul et al. (2013) [116]	De-ionized Water and Al_2O_3 coated surface (< 50 nm)	Rectangle channel	Hydraulic diameter: 0.672 mm	Heat flux of 550 kW/m^2	Experiment	CHF increased by 39 % using nanofluids, but the heat transfer coefficient decreased.
Vafaei & Wen (2014) [83]	De-ionized Water and Al_2O_3-water (25 nm, 0.001, 0.01 and 0.1 vol%)	Stainless steel tube	Wall thickness: 0.16 mm Internal diameter: 0.51 mm Length: 306 mm	Fluid temperature up to 150°C	Experiment & Simulation	CHF value increased with increasing mass flux, initial subcooling and nanoparticle concentration. Experimental testing of the new model revealed 24 % and 30 % mean absolute errors for water and alumina nanofluid, respectively.
Duursma & Sefiane (2015) [112]	Al_2O_3-ethanol (< 50 nm, 0.01, 0.05 and 0.1 vol %)	Rectangle channel	Hydraulic diameter: 0.571 mm	Heat fluxes ranged from 1.5 kW/m^2 to 9 kW/m^2	Experiment & Simulation	In the case of nanoparticle concentrations of 0.05 %, there was a maximum increment in heat transfer.
Edel & Mukherjee (2015) [105]	De-ionized Water and Al_2O_3-water (0.001 vol%)	Rectangle channel	Hydraulic diameter: 0.41 mm	Inlet temperature of 63°C and two heat fluxes of 130 kW/m^2 and 300 kW/m^2	Experiment	Using nanoparticles could stabilize bubble growth and enhance heat transfer. The duration of channel dryout decreased by 30 %.
Yu&Sur et al. (2015) [96]	Pure water and Al_2O_3-water (0.01 and 0.1 vol%)	Tube	Wall thickness: 0.25 mm Internal diameter: 1.09 mm Length: 306 mm	Inlet temperatures ranged from 80.4 °C to 90.6 °C and the maximum heat flux of 406 kW/m^2	Experiment	ONB and OFI may be delayed and suppressed by nanoparticles.

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Table 5 (*continued*)

Researchers	Working Fluids	Channel Shape	Channel Size	Heating Conditions	Analyse Methods	Main findings and/or results
Zhang et al. (2017) [117]	De-ionized Water and Graphene Oxide-water (0.05 and 0.005 wt %)	Rectangle channel	Length: 3 mm Width: 2.5 mm Height: 0.5 mm	/	Experiment	The nanoparticles increased CHF but negatively affecting heat transfer.
Moreira & Do Nascimento et al. (2017) [118]	De-ionized Water, Al_2O_3-water (20–80 nm, 0.1 vol%), Cu-water (25 nm, 0.1 vol%) and SiO_2-water (15 nm, 80 nm, 0.1 vol%)	Stainless steel tube	Wall thickness: 0.16 mm Internal diameter: 0.185 mm Length: 200 mm	Outlet temperature of 102 °C and heat flux of 350 kW/m^2	Experiment	The size of nanoparticles affected the heat transfer rate. As vapor quality increased, the heat transfer coefficient initially decreased until it reached a minimum value.
Karimzadehkhouei & Sezen et al. (2017) [119]	Pure water and γ-Al_2O_3-water (10–30 nm, 0.01, 0.2, 0.5, 1 and 1.5 wt%)	Stainless steel tube	Wall thickness: 0.1075 mm Internal diameter: 0.52 mm Length: 200 mm	Heat fluxes ranged from 270 kW/m^2 to 700 kW/m^2	Experiment	Disturbances in the stability of suspended nanoparticles and nanoparticle deposition were responsible for the deterioration in heat transfer. Subcooled boiling heat transfer deteriorated down to 12 % at the highest mass fraction.
Zhou & Luo et al. (2017) [120]	De-ionized Water and Al_2O_3-water (28 nm, 0.2, 0.5 and 1.0 wt%)	Rectangle channel	Cross section 0.32 × 2 mm, 0.6 × 2 mm, 2 × 2 mm	Heat fluxes ranged from 18 kW/m^2 to 200 kW/m^2	Experiment & Simulation	The heat transfer coefficient correlation of saturated flow boiling was proposed, and the mean relative error of the correlation was 12.3 %. Capillary effects were found to increase with decreases in hydraulic diameter.
Majid & Hossein et al. (2019) [121]	Cu-Ar	Rectangle channel	10,000 × 5000 × 5000 (Å)^3	External driving force of 0.002 eV/Å	Simulation	The addition of nanoparticles could increase heat transfer, and velocity increased by 17.8 %, 32.6 %, and 52.8 %; however, this method is not cost-effective for practical applications.
Peng & Majid et al. (2019) [122]	Cu-Ar	Rectangle channel	10,000 × 5000 × 5000 (Å)^3	External driving force 0.002 eV/Å	Simulation	Nanoparticles enhanced heat transfer and phase change, and nanoparticles had a significant impact on density fluctuations.
Dong & Jiang (2019) [123]	Al_2O_3-water (13 nm, 0.03, 0.05, 0.07 and 0.1 vol%)	Swirl microchannel	Hydraulic diameter: 0.53 mm, 0.6 mm and 0.64 mm	DC power from 200 W to 450 W	Experiment	As the vapor quality ranged between 0.4 % and 1.0 %, the heat transfer coefficient varied slightly.
Do Nascimento & Moreira et al. (2019) [82]	De-ionized Water, Al_2O_3-water (10 nm, 0.01 and 0.1 vol%) and SiO_2-water (80 nm, 0.01 and 0.1 vol%)	Tube	Wall thickness: 0.1 mm Internal diameter: 1.1 mm Length: 333 mm	Fluid temperature of 130°C	Experiment	Periodical liquid lumps were associated with saturated CHF.
Soleimani et al. (2020) [48]	Al_2O_3-water (36 nm, 1, 2 and 4 vol%)	Rectangle channel	Length: 10 mm Width: 0.1234 mm Height: 0.3049 mm	Fluid temperature of 167°C	Simulation	Heat transfer may be enhanced by adding nanoparticles. Nanofluids with a 4 vol% enhanced the overall heat transfer coefficient by up to 3 %.
Dehkordi & Karimipour et al. (2020) [124]	Fe_3O_4-water	/	/	Fluid temperature up to 426°C	Simulation	Reaching boiling time of nanofluids was shortened by enhancing the external electric field.
Dehkordi & Karimipour et al. (2020) [125]	Fe_3O_4-water	/	/	Fluid temperature up to 426°C	Simulation	Reaching boiling time of the Au microchannel was shortened by increasing the external electric field.
Yin & Hu et al. (2020) [24]	Cu-Ar	Rectangle channel	Length: 37 nm Width: 7.59 nm Height: 7.59 nm	Fluid temperatures of 150 K, 160 K and 170 K	Simulation	For nanofluids, heat transfer was enhanced, and flow boiling increased more rapidly than pool boiling.
Peng & Zarringhalam et al. (2020) [126]	Cu-Ar	Rectangle channel	10,000 × 5000 × 5000 (Å)^3	Fluid temperature of 108 K	Simulation	In practical applications, smooth surfaces were considered to be wasteful.
Husain et al. (2021) [127]	Al_2O_3-water (0–4 vol%)	Tube	Length: 975 mm Internal diameter: 3 mm	Subcooling 58.2 K	Simulation	Nanoparticle concentrations of up to 0.5 % were recommended for maximum heat transfer.
Mukherjee et al. (2021) [128]	Al_2O_3-water (20–30 nm, 0.01, 0.05, 0.1, 0.5 and 1.0 wt%)	Tube	Wall thickness: 2 mm Length: 720 mm Outer	Heat flux of 37.6–50.4 kW/m^2	Experiment	Nanofluids showed a maximum of 94 % enhancement thermal efficiency.

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Table 5 (*continued*)

Researchers	Working Fluids	Channel Shape	Channel Size	Heating Conditions	Analyse Methods	Main findings and/or results
			diameter: 45 mm			
Sharifi et al. (2022) [129]	Al_2O_3-water (20 nm, 0.05, 0.1 and 0.2 vol%)	Tube	Wall thickness: 0.5 mm Length: 800 mm Internal diameter: 26 mm	Heat flux of 50–70 kW/m^2	Experiment & Simulation	By adding nanoparticles, heat transfer and were increased by the maximum of 73 % in flow boiling.
Zhang et al. (2022) [130]	Al_2O_3-R141b (20 nm, 0.01, 0.05, 0.1 and 0.2 wt%)	Rectangle channel	Width: 2 mm Height: 2 mm	Heat flux of 4.92–24.6 kW/m^2	Experiment	Nanofluid had a greater heat transfer coefficient than R141b by 80 %.
Joshi et. al. (2023) [131]	GO-POE-R134a (0.02, 0.05 and 0.07 wt%)	Tube	Wall thickness: 1.3 mm Internal diameter: 10 mm	/	Experiment	The flow boiling heat transfer coefficient increased by 17.3 % when a nanosuspension was used at 0.07 wt%.
El Jery et. al. (2023) [132]	Al_2O_3-water (30 nm, 0.0935, 0.28 and 0.561 vol%) and CuO-water (40 nm, 0.0935, 0.28 and 0.561 vol%)	Tube	/	Heat flux of 500–4000 kW/m^2	Simulation	The difference between the average volume fractions of vapor at the outlet was 16.13 % when the nanoparticles had a volume fraction of 0.28 %.

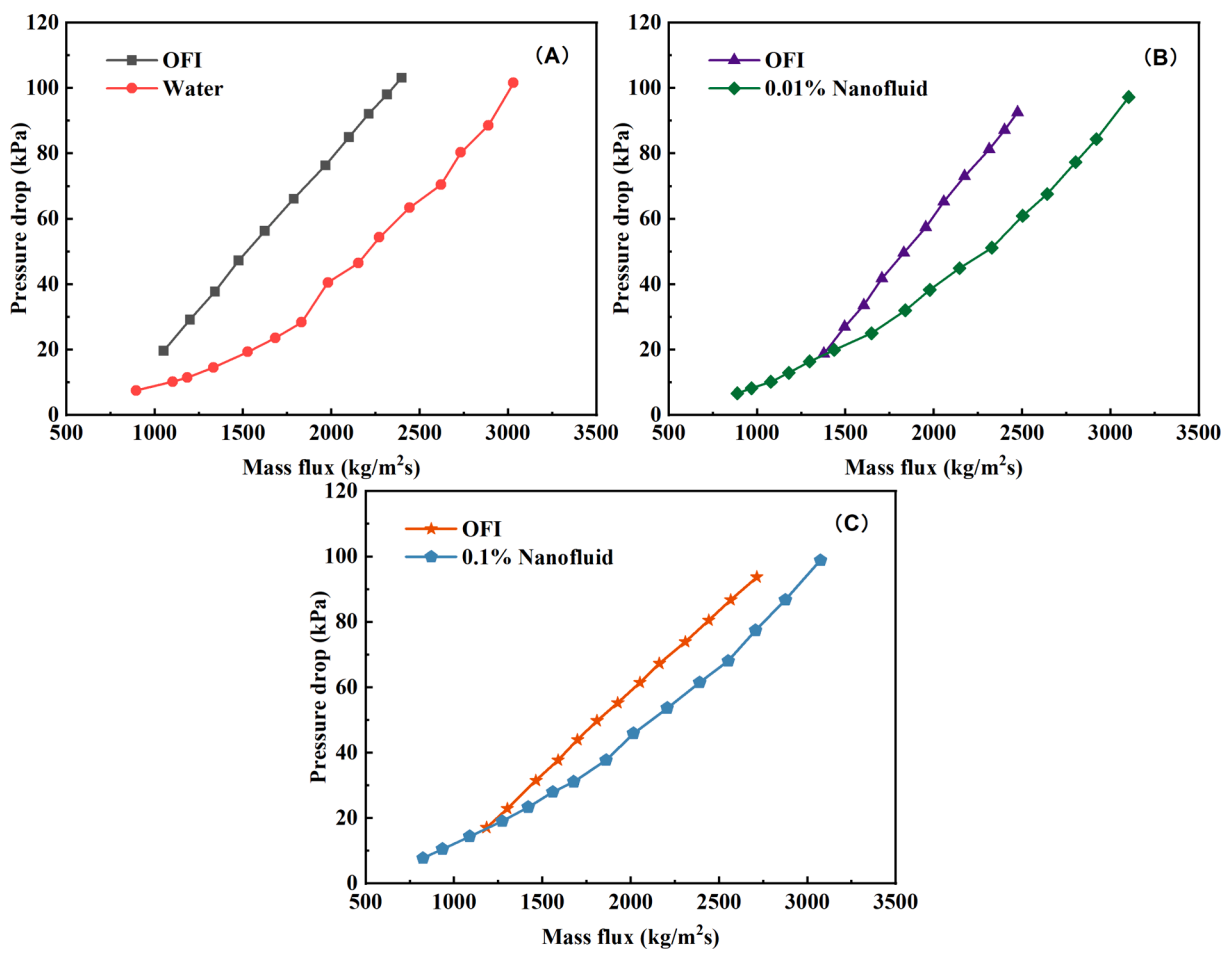

Fig. 10. Flow boiling characteristics at a heat flux of 0 W/cm^2: (A) water; (B) 0.01 vol% nanofluid; and (C) 0.1 vol% nanofluid.

Sharifi et al. [129] found that heat transfer increased in both single-phase and two-phase flows, however, the effect was more pronounced in single-phase flows. Duursma and Sefiane et al. [112] showed that the nanoparticle concentration of 0.05 % was the maximum improvement in the heat transfer using Al_2O_3-ethanol. The heat transfer rate of nanofluids was about five times higher than pure ethanol. In a simulation, Majid and Hossein et al. [121] found that the heat transfer rate increased when copper nanoparticles were added to the Argon base fluid. However, adding nanoparticles was not always cost-effective for applications.

A number of studies hold opposing views concerning the influence of nanofluids on heat transfer rates, and the majority of these results are based on experimental data. Zhang et al. [117] considered that nanoparticles reduced heat transfer rate. Simultaneously, a large number of nanoparticles could reduce heat transfer. The reason is that nanoparticle deposition blocked nucleation sites, inhibiting the heat transfer. It should be noted, however, that these conclusions were obtained using graphene oxide nanofluid and a larger channel size. It is necessary to conduct further research to determine whether the difference is related to the type of nanofluids and the size of the microchannels. Zhou and Luo et al. [120] indicated that the capillary effect increased with the decrease of hydraulic diameter as compared to different cross-sections of microchannels. Morshed and Paul et al. [116] are also in agreement with this opinion. As shown in Fig. 11, data from Morshed and Paul et al. [116] found that the heat transfer coefficient for the coated surface was lower than that of the bare surface in both single- and two-phase systems. The experiment, however, did not involve nanofluids. Several reasons can be attributed to the decrease in heat transfer, including nanoparticle deposition, nanoparticle migration, and microchannel blockage. Further study on the relationship between nanofluids and heat transfer would be beneficial if these problems could be avoided in the experiments.

Additionally, some other opinions suggest that nanoparticle diameter and concentration influence heat transfer rates. Moreira and Do Nascimento et al. [118] compared nanofluids with pure fluid in the flow boiling and found that nanoparticle concentrations of 0.1 % showed a decreasing trend in heat transfer. Despite concentrations of 0.01 % and 0.001 %, the changing trend was dependent on the diameter of the nanoparticles. It was found that small nanoparticles (Al_2O_3 20–30 nm, Cu 25 nm and SiO_2 15 nm) had decreasing trends in heat transfer, while large nanoparticles (Al_2O_3 40–80 nm and SiO_2 80 nm) had uncertainty trends. Karimzadehkhouei and Sezen et al. [119] found that heat transfer was correlated with nanoparticle concentrations of γ-Al_2O_3-water in the subcooled boiling solution. At low concentrations (0.05 wt % and 0.2 wt%), heat transfer was almost equal to that of pure water, whereas, at high concentrations (0.5 wt%, 1 wt% and 1.5 wt%), heat transfer deteriorated to about 12 %. There was an easy agglomeration of particles and nanoparticle deposition at high concentrations, which caused heat transfer degradation. Another explanation was that the hydrophilic nanoparticle deposition reduced nucleation sites in the high concentrations. Furthermore, Yin and Hu et al. [24] simulated that the low heating temperature and flow velocity have a significant effect on flow boiling heat transfer.

4.3. Pressure drop

As with heat transfer rate, no consistent conclusions have been reached concerning nanofluid influence on pressure drop. Boudouh and Gualous et al. [114] believed that Cu-water nanofluids had a higher pressure drop than pure fluid in flow boiling. However, Xu et al. [92] reported that Al_2O_3-water nanofluids reduced the pressure drop in the microchannel when flow boiling was conducted. There was a change in the flow regime from an elongated bubble to a miniature bubble, which resulted in a decrease in pressure drop and stabilization of the boiling flow. Additionally, nanofluids were located between the heating surface and the vapour phase, which resulted in the evaporation of the thin liquid film, improving heat transfer.

The heat transfer rate and pressure drop are two of the most important factors affecting the feasibility of nanofluid-microchannel systems. However, there are no evaluation criteria that take into account the economic efficiency of applications. Since some research results are different and opposing, it is difficult to choose the most appropriate nanofluid and microchannel system for application in industry. Consequently, more experiments are needed to determine the flow boiling characteristics of nanofluids in microchannel heat exchangers.

4.4. Correlation equations

There have been numerous correlations of the pool boiling or flow boiling of nanofluids in the normal channel size [133]. Nonetheless, the flow boiling of nanofluids in the microchannel is still an exploratory process. A few new correlations have been proposed, primarily based on experimental data fitting. Its mechanism is unclear. Researchers have found two main correlations between the heat transfer coefficient and the CHF, which takes into account the mechanism.

As can be seen in Fig. 12, the saturated flow boiling of heat transfer for nanofluids in the minichannel is illustrated. Based on the mechanism of nanofluids flow boiling, Zhou and Luo et al. [120] concluded that heat flux included convective boiling heat transfer and saturated nucleate boiling heat transfer. These correlations took into account the Brownian motion of nanoparticles. The total heat flux q_{tp} for saturated flow boiling, as follows:

$$q_{tp} = q_{cb} + q_{nb} = h_{tp}(T_w - T_s) \tag{10}$$

Where q_{cb} heat flux for convective boiling heat transfer; q_{nb} heat flux for nucleate boiling heat transfer; T_w channel wall temperature; T_s saturation temperature of the channel fluid. The overall heat transfer coefficient h_{tp}, as follows:

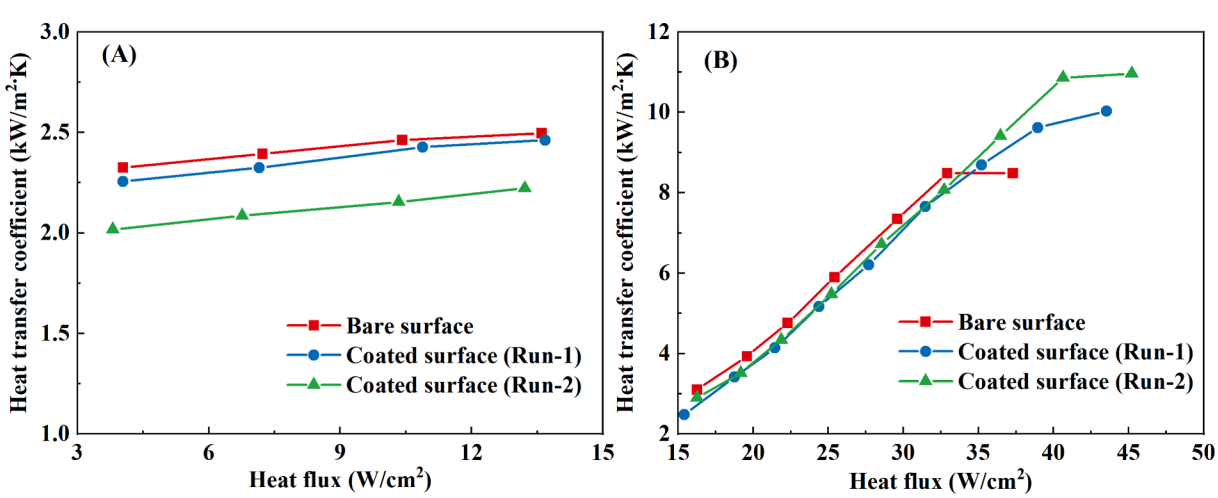

Fig. 11. Average heat transfer coefficient for the nanoparticle-coated and bare surface microchannels: (A) single-phase HTC; and (B) two-phase HTC.

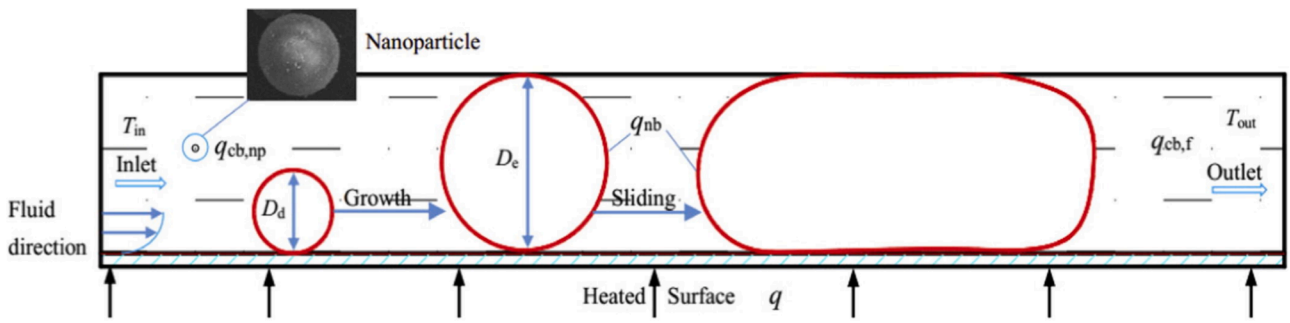

Fig. 12. An illustration of heat transfer in a minichannel under saturated flow boiling [120]. (Reprinted with permission from Elsevier).

$$h_{tp} = \frac{k_l\left[k_{np} + 2k_l - 2\varphi\left(k_l - k_{np}\right)\right]\mathrm{Nu}}{\left[k_{np} + 2k_f + \varphi\left(k_f - k_{np}\right)\right]} \times \frac{\left(K_1^{2-d_f} + K_1^{d_f} - 2K_1\right)\left(d_f - 2\right)d_f}{\left(K_1^{2-d_f} - 1\right)\left(d_f - 1\right)^2\left(\overline{D}_{np}\right)} + \left[5.2\left(\mathrm{Bo}\frac{P_\mathrm{H}}{P_\mathrm{F}}\right)0.08\mathrm{We}_l^{-0.54} + 3.5\left(\frac{1}{X_{\mathrm{tt}}}\right)^{0.99}\left(\frac{\rho_g}{\rho_l}\right)^{0.25}\right]\left(0.023Re_l^{0.8}\mathrm{Pr}_l^{0.4}\frac{k_l}{D_h}\right)$$
$$+ 0.06506\left(\frac{G(\xi)}{\xi}\right)^{0.5}\left(\frac{a_1 q}{\sigma_{lv}}\right)^{\frac{1}{2}}\frac{(Ja^*)^{\frac{5}{8}}D_s^{\frac{1}{4}}}{\mathrm{Pr}_1^{\frac{7}{4}8}}\frac{H_{lv}\rho_v}{T_s} \tag{11}$$

Where k_l is fluid thermal conductivity; k_{np} is nanoparticles thermal conductivity; and Nusselt number is described as:

$$\mathrm{Nu} = 2 + \left(0.4\mathrm{Re}_{l,\mathrm{np}}^{1/2} + 0.06\mathrm{Re}_{l,\mathrm{np}}^{2/3}\right)\mathrm{Pr}_l^{0.4}\left(\frac{\mu_\infty}{\mu_\mathrm{w}}\right)^{1/4} \tag{12}$$

Where Re is Reynolds number; and Pr is Prandtl number.

The coefficients are shown as:

$$K_1 = D_{np,min}/D_{np,max} \tag{13}$$

$$d_\mathrm{f} = 3 - \frac{\ln\varphi}{\ln\left(D_{\mathrm{np},min}/D_{\mathrm{np},max}\right)} \tag{14}$$

Lockhart-Martinelli parameter X_{tt} is shown as:

$$X_{\mathrm{tt}} = \left(\frac{\mu_l}{\mu_\mathrm{g}}\right)^{0.1}\left(\frac{1-x}{x}\right)^{0.9}\left(\frac{\rho_\mathrm{g}}{\rho_l}\right)^{0.5} \tag{15}$$

In addition, Ahn and Kang et al. [88,89] proposed a new flow boiling model based on the wetting-zone fraction. In this case, the total heat transfer was still equal to the force convection plus nucleate boiling. As the surface wettability affected CHF in the flow boiling, the nucleate boiling area fraction was considered. The overall heat transfer coefficient h_{tp} for saturated flow boiling, as follows:

$$h_{\mathrm{tp}} = A_{fc}h_{fc} + A_{nb}h_{nb} \tag{16}$$

$$A_{fc} + A_{nb} = 1 \tag{17}$$

Where A_{fc} area fraction of forced convection; and A_{fc} area fraction of nucleate boiling.

$$N_a = C\left(\frac{1}{r^*}\right)^m\left(1 - \cos\theta_a\right) \tag{18}$$

Where the critical radius of cavity r^* is shown as:

$$r^* = \frac{2T_{\mathrm{sat}}\sigma}{h_{\mathrm{fg}}\rho_g\Delta T_w} \tag{19}$$

$$D_d = 0.626977 \times \frac{(2 + 3\cos\theta - \cos^3\theta)}{4} \times \left(\frac{\sigma}{g(\rho_L - \rho_V)}\right)^{1/2} \tag{20}$$

$$A_{nb} \sim N_a \times D_d^2 \tag{21}$$

Besides these two correlations, the majority of correlation equations for CHF are empirical formulas that are fitted using experimental data [56,83,134]. The influence factors of CHF were therefore limited to a few variables. In order to solve these problems, DolatiAsl and Bakhshan et al. [135] attempted to consider comprehensive factors and suggested several CHF correlations, as demonstrated in Table 6. Although CHF for subcooled nanofluids can now be predicted, few formulas have been proposed based on the mechanism, which still requires further research.

5. Conclusion and future work

The flow boiling of nanofluids in microchannel heat exchangers has been investigated, with the consideration of increasing density of packaging and the need for heat dissipation. This article summarized some of the most significant results over the past 15 years, of which the topic is controversial and has many opposing views. Therefore, some key conclusions and suggestions can be drawn as follows:

Table 6
CHF models for subcooled nanofluid flow in the microchannel [135].

Model	Nanoparticles	Correlation
1	Al_2O_3	$\mathrm{CHF}=\rho_{np}^{0.000222} \times k_{np}^{-0.005021} \times G^{0.637622} \times T_{in}^{1.217939} \times L^{0.251812} \times D^{0.778025} \times \rho_l^{3.326245} \times \rho_v^{2.267713} \times \varphi^{0.017829} \times P^{-1.952992} \times 0.000004629 - 253.57$
2	ZnO	$\mathrm{CHF}=(\rho_{np}^{0.009097} \times k_{np}^{-0.004792} \times G^{0.635333} \times T_{in}^{1.113257} \times L^{-0.261948} \times D^{0.775767} \times \rho_l^{1.910724} \times \rho_v^{2.090967} \times \varphi^{0.017941} \times P^{-1.780308} \times 0.055314 - 250.53) \times We_d^{0.121}$
3	Diamond	$\mathrm{CHF}=(N_n^{-0.056164} \times \mathrm{G}^{0.673573} \times T_{\mathrm{in}}^{0.956687} \times L^{-0.235763} \times D^{0.838306} \times \rho_l^{3.186126} \times \rho_v^{2.259023} \times \varphi^{0.018221} \times P^{-1.950025} \times 0.000057 - 184.28) \times We_d^{0.121}$
4	GO	$\mathrm{CHF}=N_n^{-0.046604} \times G^{0.799146} \times T_{\mathrm{in}}^{-0.066666} \times L^{0.336549} \times D^{0.050885} \times \rho_l^{-0.056065} \times \rho_v^{0.776294} \times \varphi^{0.036149} \times P^{-0.084568} \times 4.889356 + 43.54$

- Most recent experiments are being conducted on various channel sizes. But it was suggested that the standard (or the same) criteria should be applied during the microchannel experiments, in order to make the experimental results more reliable, comparable, and accurate. The transition criteria between macro-/mini-channels and microchannels can be described as C_o 1, which indicates that the area of bubbles is similar to the cross-sectional area of the channel.
- Further improvements need to be made to the transition criteria since all criteria are currently created based on pure fluid coolants. There is a need to verify whether nanofluids as coolants will alter the results due to the agglomeration and suspension of nanoparticles. Also, the current criteria only consider the parameters of surface tension force, hydraulic diameter, and phase transition, and they also can be affected by the conditions, including channel structure, surface situation, and type of coolant, which have not yet been investigated. Finally, a transition region needs to be identified in order to understand the whole process of channel transition during the phase change, which possibly can be implemented through numerical analysis or physical testing in this area. Since the rules change with various flowing states, it is necessary to divide the flowing boiling of nanofluids in microchannels into subcooled boiling and saturated boiling, in order to achieve accurate results.
- CHF is regarded as a critical evaluation parameter. There is a fierce debate regarding the mechanism and changing trends of CHF since flow boiling is highly sensitive, leading to a different result even when the experiment procedure and working conditions are all the same. Therefore, a large number of experiments and uncertainty analyses are required. As an example, thermocouples should be placed within the internal channel rather than on the outside wall in order to reduce thermal inertia and thermocouple hot junctions. The study of measurement methods is also beneficial to improve the uncertainty at the same time. During the experiments, periodic liquid lumps, thermal instabilities, and backflows in the microchannel can be tested.
- In comparison to pure fluids, nanofluids can reduce temperature fluctuations, improving the safety of modern devices. In the future, it is recommended that the pressure fluctuation can be compared with the temperature fluctuation. In most experiments, Al_2O_3 is used as a nanoparticle. Various nanofluids should be compared in order to determine the most suitable nanofluid for flow boiling in microchannel heat exchangers, and using more could help explain flow boiling behaviours. In addition, new nanofluids (hybrid nanofluids and magnetic nanofluids) can also be taken into account.
- In cooling systems, both the heat transfer rate and the pressure drop are considered significant parameters, while the results of the present experiments are still controversial. To resolve the problem fundamentally, it is necessary to clarify the mechanism issues first, followed by improving the experimental accuracy. It is important to note that the criteria for evaluating an application must be established, taking into account both the cost and complexity of the application. In the present study, most of the correlations between CHF and heat transfer rate are generated with the fitting method using experimental data. It was also suggested to consider other parameters and mechanisms from the working principles of the applications, in order to derive a more accurate correlation.

CRediT authorship contribution statement

Meng Wang: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Ji Wang:** Writing – review & editing, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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